

Phishing Attack Simulation & Awareness

Project 3 - Cyber Security & Ethical Hacking

CODEVEDX Internship Program

Project Title: Phishing Attack Simulation & Awareness

Organization: CODEVEDX

Date: June 02, 2026

Purpose: Educational - Awareness Training

1. Introduction

Phishing is one of the most prevalent cyberattack vectors. 91% of all cyberattacks begin with a phishing email. This project simulates phishing attacks in a controlled educational environment to demonstrate how they work and how to prevent them.

Objectives

- Understand phishing attack mechanics
- Create realistic simulation pages
- Develop phishing detection tools
- Build interactive awareness training
- Generate analytical reports

2. Phishing Overview

Types of Phishing

Email Phishing

Mass emails impersonating companies.

Spear Phishing

Targeted attacks using personal info.

Whaling

Targeting executives.

Clone Phishing

Copies of legitimate emails.

Smishing

Phishing via SMS.

Vishing

Phishing via phone calls.

Statistics

- 91% of attacks start with phishing
- 3.4B phishing emails daily
- Average breach cost: \$4.76M

3. Project Architecture

Technology	Purpose
Python 3	Core language
Flask	Web framework
HTML/CSS	Templates
JSON	Data storage

4. Implementation

The project uses a Flask web server hosting 3 phishing scenarios. Each scenario captures user interactions and displays immediate awareness warnings. Passwords are masked for safety.

Key Features:

- 3 phishing scenarios
- Real-time logging
- Instant awareness warnings
- JSON API reports
- Password masking

5. Simulation Scenarios

Scenario 1: Fake Google Login

Attribute	Value
Attack Type	Credential Harvesting
Risk Level	HIGH

- URL is not accounts.google.com
- Hosted on different domain
- No Google OAuth

Scen

Attribute	Value
Attack Type	Credential Harvesting
Risk Level	CRITICAL

- False urgency alert
- Fake bank name
- Fake SSL badges

Scen

Attribute	Value
Attack Type	Urgency Scam
Risk Level	HIGH

- Generic "Dear Customer" greeting
- 24-hour suspension threat
- Password request via email

6. Detection Tools

URL Checker

Analyzes URLs for 9 phishing indicators including HTTPS, typosquatting, suspicious TLDs, and phishing keywords.

Email Analyzer

Analyzes email content across 7 categories: urgency, threats, greed, credential requests, impersonation, suspicious links, and grammar.

7. Awareness Training

4 interactive modules with quizzes covering phishing identification, types, and protection strategies.

8. Results & Analysis

The simulation tracks page visits, credential submissions, awareness warnings viewed, and unique IPs. All data is logged to JSON for analysis.

9. Recommendations

For Individuals

- Enable 2FA on all accounts
 - Use a password manager
 - Verify URLs
 - Never click suspicious links
 - Be suspicious of urgency
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- Quarterly phishing training
 - Implement SPF/DKIM/DMARC
 - Deploy email filtering
 - Maintain incident response plan

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10. Conclusion

This project demonstrates phishing attack mechanisms through realistic simulations and provides tools for detection and awareness training. The key takeaway is that phishing prevention requires technical controls, human awareness, and organizational policies working together.

11. References

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- [6] CISA Phishing Guidance
- [7] Microsoft Security Report
- [8] Google Safe Browsing
- [9] Flask Documentation
- [10] Python Documentation